



Multi-Level Classifier for Medicinal Plant Type and Health Prediction

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ABSTRACT

Plants fulfill the food, shelter, fuel and other life related requirement from ancient time. Human is continuously performing a lot of efforts to overcome the disease related to plant decays. To cure any disease it is necessary to identify whether a leaf is healthy or diseased. This work helps to identify two important attributes from an plant image i.e. plant leaf type and to predict leaf is either healthy or unhealthy. This combined phenomena is achieved by multilevel classification. Well accepted publically available SMVU Katra dataset is used. Multi-class classification paradigm is used for classification on both levels i.e. plant leaf type and plant leaf health. A brief discussion with six machine learning classifiers are presented. These six classifiers are Support Vector Machine(SVM), Native Bayes(NB), Logistic Regression(LR), Extreme Gradient Boosting (XGBoost), k-Nearest Neighbours (KNN) and Random Forest(RF).Random Forest classifier reports the best result with accuracy 86.32% and F1-Score 86.22 %. This work is presents a state of art framework for leaf type and leaf health detection.

Keywords: Medicinal Plants; Unhealthy Leaves; Machine Learning; Multilevel Classification; Computer Vision; k-fold Cross Validation; Performance Metrics; Ayurveda.

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1.0 INTRODUCTION

Plants play an important role in human life. The idea of human life without plants is meaningless. Plants fulfill all the basic needs related to human life like oxygen, food and fuel etc. Since ancient times, every part of the plant i.e. root, stem, leaf, flower and fruit etc. are being continuously used as medicine in the medical field especially in Ayurveda. With the use of which not only diseases are being successfully treated but it also had a positive effect on human health. During the Covid-19 pandemic, these medicinal plants played a vital role in

saving human lives, they have a huge impact in the medicine market of the India and the world. It has a huge impact on the GDP (Gross Domestic Product) of the India and on the income of farmers [1].

The Government of India has also emphasized on Ayurvedic medicine in its budget 2024-25[2]. The MoAYUSH is responsible for promoting education and research in Indian systems of medicine like Ayurveda, Yoga and Naturopathy and Homeopathy. The Union Budget 2023-24 allocated INR 3,6475 million to the Ministry of Ayurveda and Naturopathy (MoAYUSH),

which is a 19.6% increase from the budget in last financial year 2022-23. This indicates that medicinal farming is proving helpful in strengthening the economic condition of the country and farmers.

Due to market demand and good profits, the trend of cultivation of medicinal plants has increased among farmers.

For good yield, it is very important to have healthy plants, for this purpose it becomes essential to identify the diseased leaves of the plants at the right time.

Generally, farmers have been doing this work for a long time with their experience and periodic observations of plants. Apart from this, some ancient scientific methods are also being used to identify diseased plants, but these methods are expensive and require qualified operators and technicians.

Since most of these methods are dependent on laboratories, it is very difficult for farmers living in remote areas to use them.

But in this era of 21st century, Internet, mobile services and data storage facilities have made human life easier. It promotes the use of machine learning based applications. There has not been much work done on identify plant leaf type and leaf health simultaneously. We present an approach to predict leaf type and its health simultaneously, based on machine learning approach.

In this paper, section 2, presents a brief discussion of the different reported research strategies for leaf disease detection by using existing machine learning approaches on different plant leaves. Section 3, of the paper presents the machine learning based mechanism for plant leaf classification and detailed description of SMVU dataset and reviews various classification techniques used in detection of plant diseases. Section 4, provides a bar graph based visual representation with summary of results and classification techniques. Section 5, of this paper offers conclusions and suggestions for the future.

Main contributions of this research work are as following:-

- a. Multi-level framework for leaf type and leaf-health detection.
- b. Performance analysis of proposed framework using different metrics with existing machine learning classifiers.

2.0 RELATED WORK

In this section various major machine learning based reported works to identify healthy and unhealthy leaves of different plants are explained. In medicinal plants also like other plants, disease can occur in any part of the plant such as root, stem, leaf or panicle etc. In this section diseased leaves of the plants are considered.

Chauhan et al.[3] explained the technique that uses computer vision techniques to segment and classify leaf diseases in *Pongamia Pinnata* L. and *Jatropha Curcas* L. plants. Using colour, shape, and texture features for classification, a hybrid neural network with super pixel clustering is suggested for disease region segmentation. High specificity, sensitivity, and classification accuracy are achieved by applying various machine learning classifiers, indicating the efficacy of the suggested method in plant disease diagnosis.

Bose, et al.[4] proposed a study that suggests an automated method that uses machine learning and image processing to identify plant leaf diseases. Support Vector Machine (SVM) classifiers are utilised for classification, whereas methods such as Histogram Orientated Gradient (HOG) and Local Binary Pattern (LBP) are employed for feature extraction. High accuracy is attained by the system. But the scalability of big datasets is not addressed in the work.

Pham et al. [5] present a work, in this research work, they have demonstrated preliminary disease classification on a limited plant dataset from various available plant image datasets using feed-forward neural networks and shows promising results. This work lacks scalability tests across many other types of plant leaf datasets and uses a small dataset of 450 leaf images of mango plant.

Padol et al.[6] explained how to classify leaf diseases using linear Support Vector Machine (SVM). Images of grapes leaves are used as input and preprocessed. Clustering methods are used to detect them and extract texture and colour information from them.

Luo et al.[7] presents a following research work. In this study researchers examine the use of computer vision (CV) technology in indoor farming and urban farming. It looks at how CV applications in controlled-environment agriculture (CEA), including as disease detection, growth monitoring, and maturity categorisation, can improve agricultural output. To promote CV developments in CEA, it also examines datasets and research gaps. Practical applications, real-world difficulties, and metrics to measure the effectiveness of computer vision (CV) models are not covered in this research work. It also doesn't offer clear guidance on how to use these models in different conditions or environments.

When we performed the dataset thorough study of data and literature review, we found a lack of a model to handle both classification simultaneously, i.e leaf types prediction and leaf health prediction. Our proposed model provides a state-of-art technique to address these both predictions simultaneously.

3.0 METHODOLOGY

3.1 Model Description Diagram

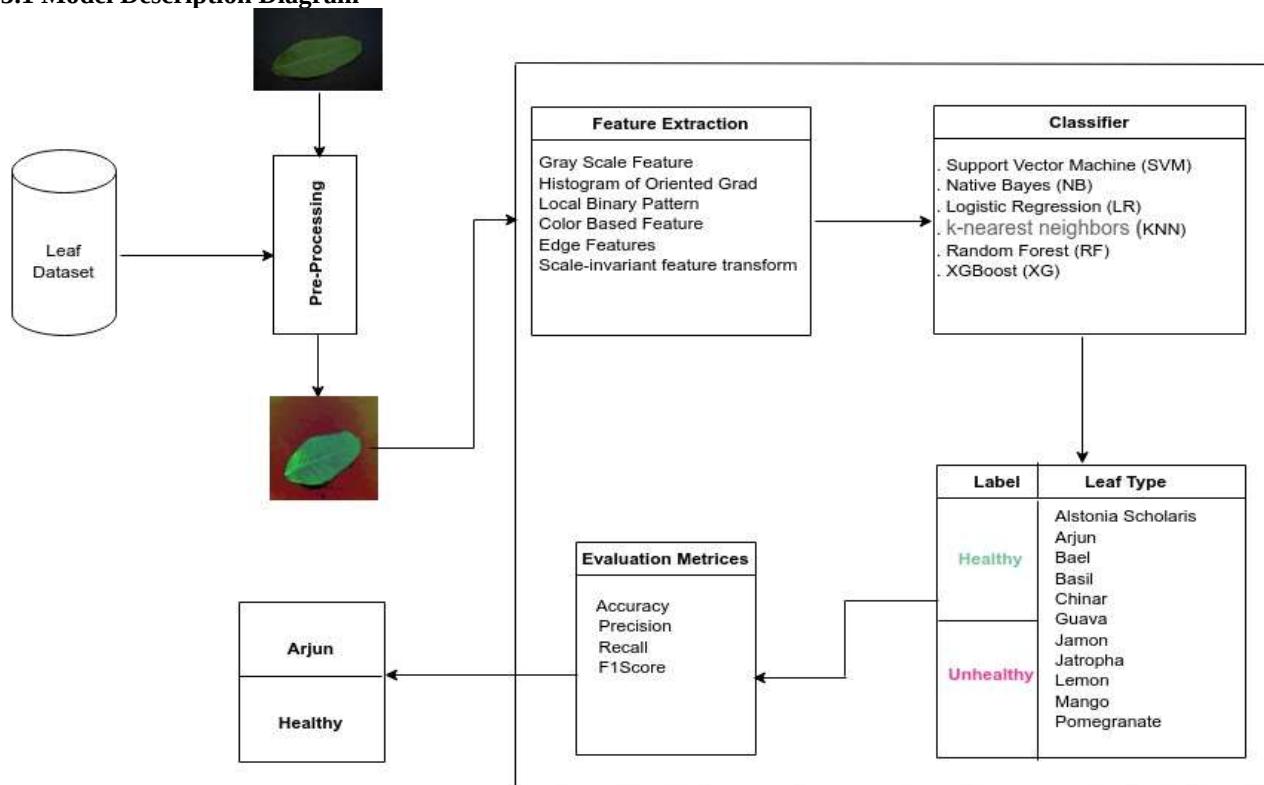


Fig 1: Multi-level Prediction Work-flow

3.2 Algorithm Description

Open source Python software (Automatic system) analyses plant images, determines if the leaf of the plants are diseased or healthy, and assesses the model's accuracy. It starts with importing the required libraries, such as scikit-learn for machine learning and OpenCV for image processing. These libraries are essential for feature extraction, to apply machine learning classifier and for evaluation metrics.

Images are read from specified location, the images are loaded when it navigates through hierarchical locations that correspond to various plant categories and their respective states of health, resized to 128x128 pixels, and labelled according to the folder structure using the *load_images_from_folders* method. The *extract_features* function is then used to flatten these images into 1D arrays, or feature vectors. LabelEncoder is used to encode the labels into numerical numbers. Due to limited amount of data set available, we used k-fold cross validation technique to make maximum use of the entire data set for both training and evaluation. It minimizes the possibility of over-fitting and generalizing to previously unseen data. Grid cv method is applied to hyper-parameter tuning of each machine learning classifier. It helps to achieve higher performance metrics with possible hyper parameter combinations.

Six classifiers SVM, NB, LR, XGBoost, KNN and RF are trained on the training set after the dataset is divided into training and testing sets. The test set is used to assess the model's performance using measures such as accuracy, precision, recall and F1-score.

The mechanism offers a feature to forecast the uses of the trained model for classification after resizing and flattening a fresh image to anticipate its health state. The output is recorded with the anticipated label.

3.3 Data set description

A well accepted dataset is used to perform experiments. Data set is made publically available by SMVU, Katra, India[19]. The Data set is made-up of 11 medicinal plant leaf images. These medicinal plants are economically beneficial for farmers and according to Indian subcontinent climate. These are Alstonia Scholaris, Arjun, Bael, Basil, Chinara, Guava, Jamon, Jatrophia, Lemon, Mango and Pomegranate. The leaves are classified in total 20 categories. In which 10 categories are of unhealthy medicinal plants and the remaining 10 categories are also marked as healthy medicinal plants. In this dataset, images of healthy leaves of Bael as well as unhealthy images of Tulsi plants are not present.

Total number of leaf images of medicinal plants in dataset are 3641. 1856 leaf images are unhealthy and 1785 images are of healthy category. Based on the ratio of number of healthy and unhealthy images, it can be said that the database is almost balanced. Balance dataset makes model training efficient and effective because it helps to prevent the model from becoming biased towards one of the classes. All these leaf images have been collected in digital image file format JPEG (*Joint Photographic Experts Group*). These images are captured with a high-resolution professional camera. After studying all online versions of the data set, version 5[20] that is Published: on January 20, 2022,

contains significantly more images and its use had a positive impact on the results obtained. Version 5 of dataset is larger than version 4. In version 5 there are 170 healthy leaf images and 93 unhealthy more leaf

images than version 4. As a result, there are now 3904 leaves total in the dataset, consisting of 1955 healthy leaves and 1949 diseased ones. it indicates a balanced dataset.

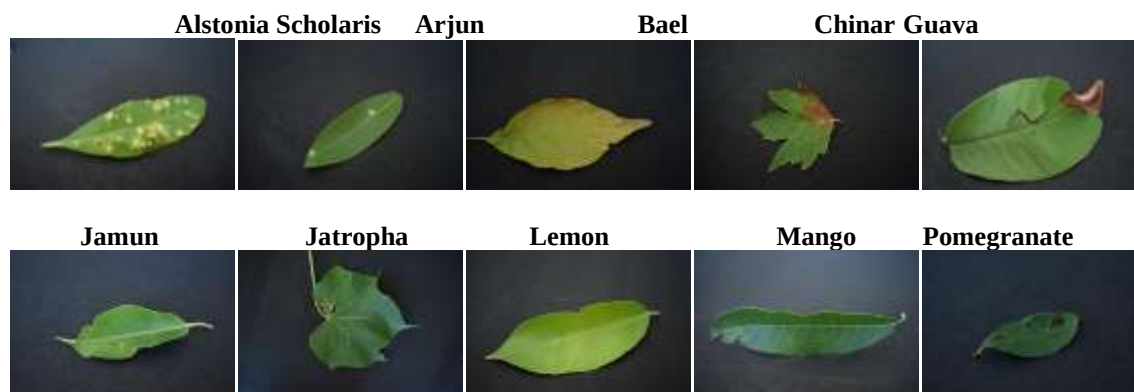


Fig 2: Diseased leaf images of various medicinal plants



Fig. 3: Healthy leaf images of various medicinal plants
<https://data.mendeley.com/datasets/compare/hb74ynkjc>

Bar chart in **Figure 4** displays the distribution of both healthy and unhealthy leaves across various plant species, with the count of healthy and unhealthy leaves for each species labelled on top of each bar. With almost 270 healthy leaves, plants like guava, jamun, and pomegranate have the highest counts. Similarly, from the unhealthy leaves point of view, jamun with

more than three hundred and mango and pomegranate with around 300 leaves are the maximum in the dataset. Species like Bael and Basil have zero healthy and unhealthy leaves respectively. This representation helps to understand overall distribution of healthy leaves and unhealthy leaves in dataset.

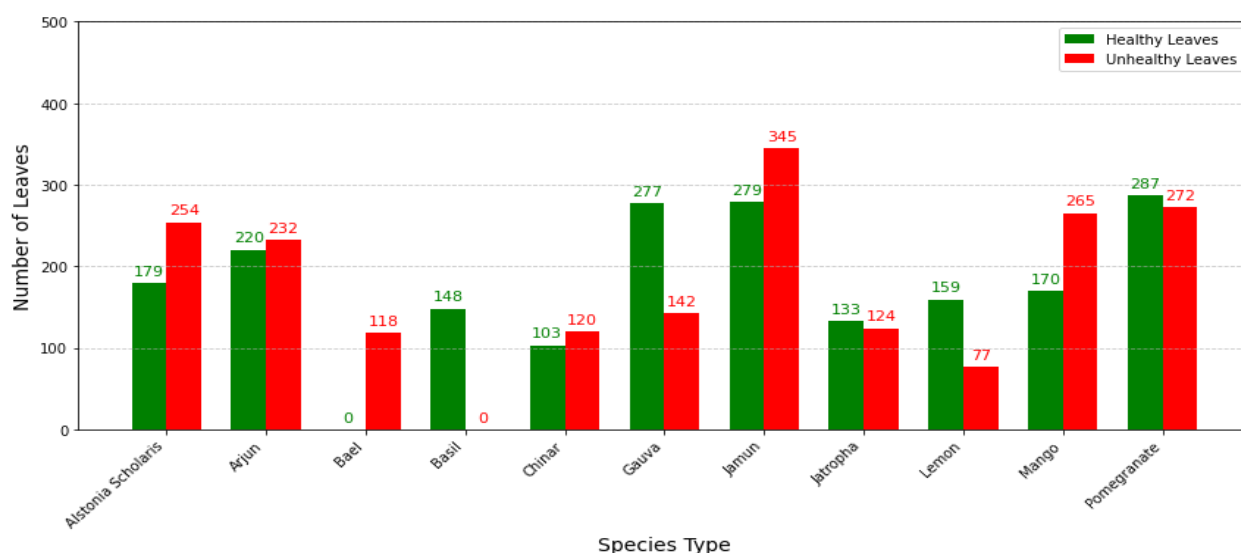


Fig. 4: Quantity of various healthy and unhealthy leaves collected in the dataset

The medicinal properties of *Alstonia Scholar*[18] is used in the treatment of malaria, jaundice, gastrointestinal problems, cancer and many other complex diseases. A total of 433 leaf images of *Alstonia Scholaris* were collected in the database, out of which 254 leaves were unhealthy while 179 leaves were healthy.

Arjuna (*Terminalia arjuna*) is widely used for treatment of cardiovascular diseases, including heart diseases and related chest pain, high blood pressure and high cholesterol[8]. The total number of leaf images of Arjuna plant in the database is 432, out of which 232 are unhealthy and 220 are of healthy leaves.

Bael (*Aegle marmelos*) fruits are used in the treatment of chronic diarrhea, dysentery, and peptic ulcers, as a laxative and to recuperate from respiratory affections in various medicines[9]. 118 unhealthy leaf images are of Bael present for analysis in the dataset. There is no image of healthy bael leaf is present.

Basil (*Ocimum basilicum*) helps to normalize blood sugar, blood pressure and lipid levels. It also reduces psychological stress through its positive effects on memory and cognitive function and its anxiolytic and antidepressant properties. It is also widely used as a hand sanitizer, mouthwash and water purifier[10]. Total 148 healthy leaves of Basil have been taken for analysis in the database. Dataset contains no image of unhealthy leaf of Basil.

Chinar (*Platanus orientalis L*) is used in dental medicine, wound healing, inflammation and stomach troubles[11]. A total of 223 leaves of chinar have been collected in the database, of which 120 are unhealthy and 103 are healthy leaf images.

Guava (*Psidium guajava*) is beneficial in Gastrointestinal infections, Oral/dental infections, Skin infections, Diabetes, Liver problems[12]. 277 healthy and 142 unhealthy leaf images have been collected in the guava dataset, that is a total number of 419 guava leaf images.

Jamun (*Syzygium cumini*) is an excellent source of bioactive components like antioxidants, iron, and

vitamin C. It serves a beneficial role against metabolic syndrome.[13]. A total of 624 leaf images of Jamun have been collected in the dataset, of which 345 are unhealthy and remaining 279 are healthy leaf images.

Medicinal studies have shown that *Jatropha Curcas* has significant activity as anti-inflammatory, antidiarrheal, antihypertensive and anticancer agents[14]. A total of 257 leaf images of *Jatropha* have been collected in the dataset, of which 124 are unhealthy and remaining 133 are healthy leaf images.

Due to its positive properties like anti-cancer, anti-tumor and anti-inflammation, lemon (*Citrus limon*) has its own importance from the medicinal point of view[15]. There are total 236 leaf images, of which 77 and 159 are unhealthy and healthy images respectively in dataset.

Mango (*Mangifera indica*) has anti-diabetic, antioxidant, antiviral, cardiogenic, blood pressure lowering, anti-inflammatory properties[16]. This dataset contains 170 healthy and 265 unhealthy mango leaves.

Pomegranate (*Punica Granatum*) is used medicinally to treat various diseases including high blood pressure, high cholesterol, oxidative stress, hyperglycemia, and inflammation[17]. A total of 559 leaf images of Pomegranate have been collected in the dataset, of which 272 are unhealthy and remaining 287 are healthy leaf images.

3.4Pre-Processing

Data pre-processing is the first step to make machine learning analysis effective. This important step positively impacts the efficiency and accuracy of any machine learning algorithm.

To provide images of a uniform input size for the models, each image are reduced to 128x128 pixels. Every image is flattened from a 2D matrix to a 1D vector. This Figure 5 is the display of the original image and pre-processed image.

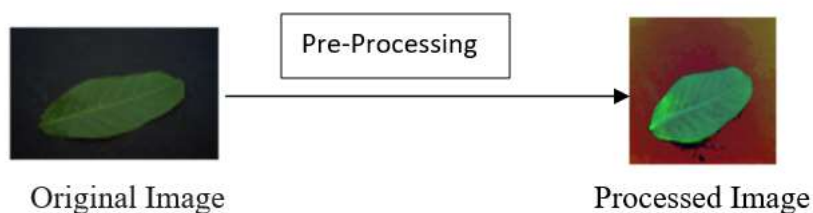


Fig.5 : Image Pre-processing Workflow

3.5Machine Learning Classifiers

A machine learning classifier is an algorithm that learns from a previously classified dataset and then attempts to predict accurately the class and label of a given unseen data set and making decisions by analyzing complex data.

Support vector machines (SVM) is powerful machine learning algorithm that is useful for regression and classification problems because it works by calculating the optimal hyper plane

in a high-dimensional space that efficiently separates different classes of data points. When dealing with data from a finite sample, it is applied.

A probabilistic machine learning algorithm called Naive Bayes(NB) is used to classify diseases of plant leaves. The underlying assumption is that, given the class name, the features that are applied to predict the class—such as distinct pixel values in leaf images—are independent of one another. The algorithm determines which class has the highest probability by calculating

the probability of each class given the input features. It was perfect for instances requiring real-time plant disease classification since it allowed for quick training[21].

A well-known machine learning classifier, logistic regression(LR) is primarily applied to binary classification issues, in which the objective is to divide data into two different categories. The logistic function is perfect for estimating probabilities since it converts any real-valued integer into a range between 0 and 1. Every characteristic is given a coefficient that represents how it affects the possibility of the outcome. Whenever probabilistic interpretation and clarity of the model are crucial, logistic regression is efficient and effective.

XGBoost (*Extreme Gradient Boosting*) is a effective machine learning classifier that has demonstrated significant efficacy in the identification of plant diseases. The gradient boosting architecture serves as its foundation, combining a group of weak learners—typically decision trees—to increase prediction accuracy. High accuracy of XGBoost makes it ideal for identifying minute variations in the health status of plants. XGBoost can detect patterns of plant diseases by learning extensive interactions in data, including

differences in colour, texture and shape in plant leaf images.

The k-Nearest Neighbours (KNN) classifier is an instance-based learning technique that does not build a generalisation model; instead, it derives its predictions on the dataset's nearest examples. It works most effectively in situations when disease symptoms on plant leaves are frequently seen as distinctive leaf patterns. KNN can reliably categorise diseases according to visual symptoms by selecting the most comparable samples, giving farmers important information to take prompt action and minimise crop damage.

Random Forest(RF) is a powerful machine learning classifier that is effective for both classification and feature selection, and being used extensively in plant disease detection due to its high accuracy, robustness, and ability to handle complex data. Random forest generates several decision trees and aggregates their outputs to produce a model that enhances generalisation and reduces the possibility of over-fitting. Random Forest classifier is useful for real-world applications because it can produce accurate predictions even when features are missing.

3.6 Performance Metrics Used

		ACTUAL VALUES	
		Positive (+VE)	Negative (-VE)
PREDICTED VALUES	(+VE)	TP	FP
	(-VE)	FN	TN

Fig. 6: Confusion Matrix

Abbreviations:

True Positives (TP) - When the positive class was accurately predicted by the model.

True Negatives (TN) - When the negative class was accurately predicted by the model.

False Positives (FP)-When the positive class was incorrectly predicted by the model, but the actual class was negative.

False Negatives (FN) -When the negative class was incorrectly predicted by the model but the actual class was positive.

Performance metrics play an vital role to evaluate the efficiency of any machine learning model. The mathematical equations for these metrics are given below.

Accuracy:- Accuracy is the one of the most fundamental and extensively used predictions for assessing a classification model's performance. It measures the proportion of correct predictions out of the total predictions, both positive and negative. It shows the percentage of plant leaves that were correctly classified as either healthy or unhealthy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:-Accuracy has its limitations, but the precision metric helps deal with them. The percentage of positive predictions that were actually correct is measured by precision. It emphasizes the accuracy of the positive predictions. The following equation reflects how many of the leaves predicted as unhealthy are actually unhealthy.

$$Precision = \frac{TP}{TP + FP}$$

Recall:- It is used to determine how well a classification model detects all of the real positive cases. It indicates the model's efficiency to correctly detect unhealthy leaves.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:- The F1-score represents a balanced metric that takes into account both false positives and false negatives and useful when false positives and false negatives are equally important. It is a helpful method for evaluating a classification model's overall efficiency.

$$F1 - Score = \frac{2 * TP}{2 * TP + FP + FN}$$

4.0 Results and Discussion

Bar graph show cases in **Figure 6** displays the accuracy of applied machine learning classifiers. In the bar graph x-axis represents the classifiers used like SVM, NB and LR and the y-axis displays the accuracy(%) achieved for each model. In following bar graph figure (6), RF seems to surpass the other classifiers, while NB has a significantly lower accuracy. These illustrations make quick assessment and model selection for additional study easier.

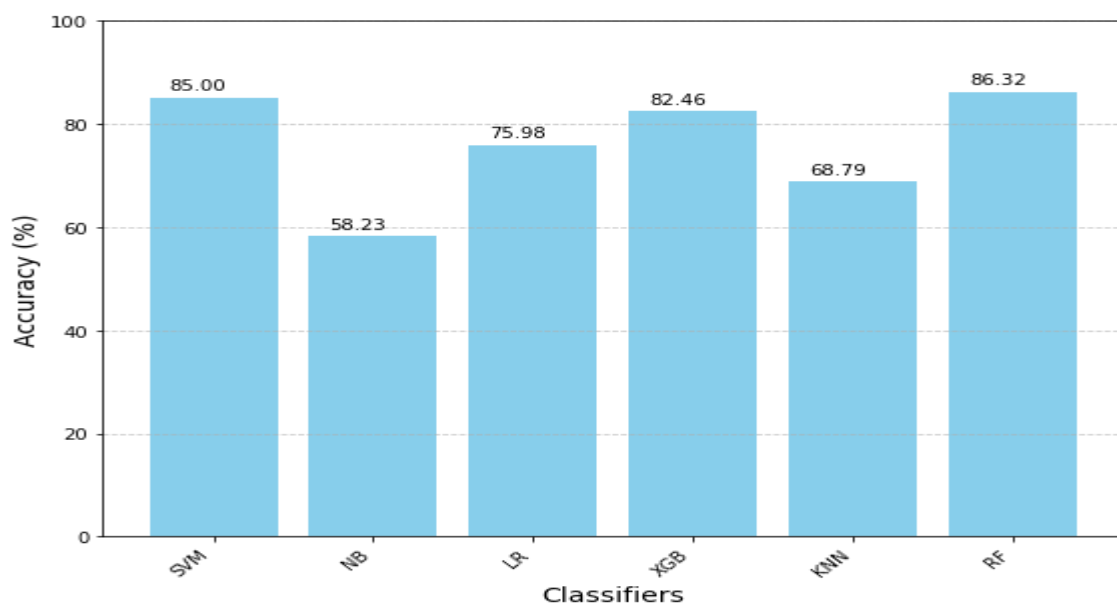


Fig. 6: Accuracy of machine learning classifiers.

Bar graph in **Figure 7** displays the precision of employed machine learning classifiers. The x-axis represents the classifiers used and the y-axis represents

the precision percentage of each mode. Random forest(RF) seems to outcome the other classifiers , while NB has a significantly lower Precision.

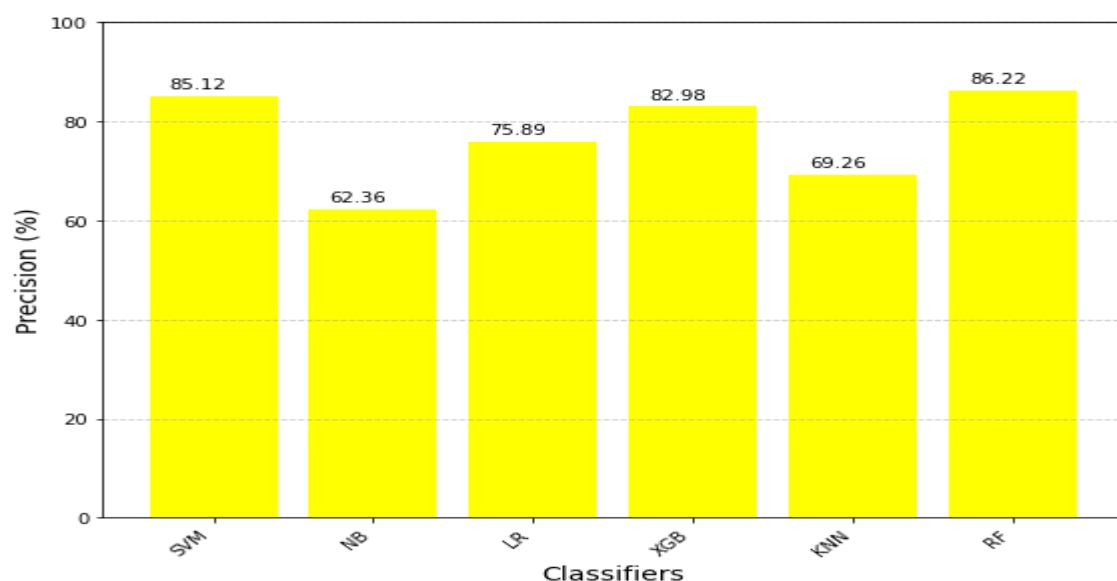


Fig. 7: Precision of machine learning classifiers.

The bar graph shown in **Figure 8** displays the recall of machine learning classifiers. In the bar graph, the x-axis represents the classifiers used and the y-axis displays the recall percentages achieved by each model.

Compared to the other classifiers, RF appears to perform better, and NB has a noticeably lower recall. These results helps in model selection.

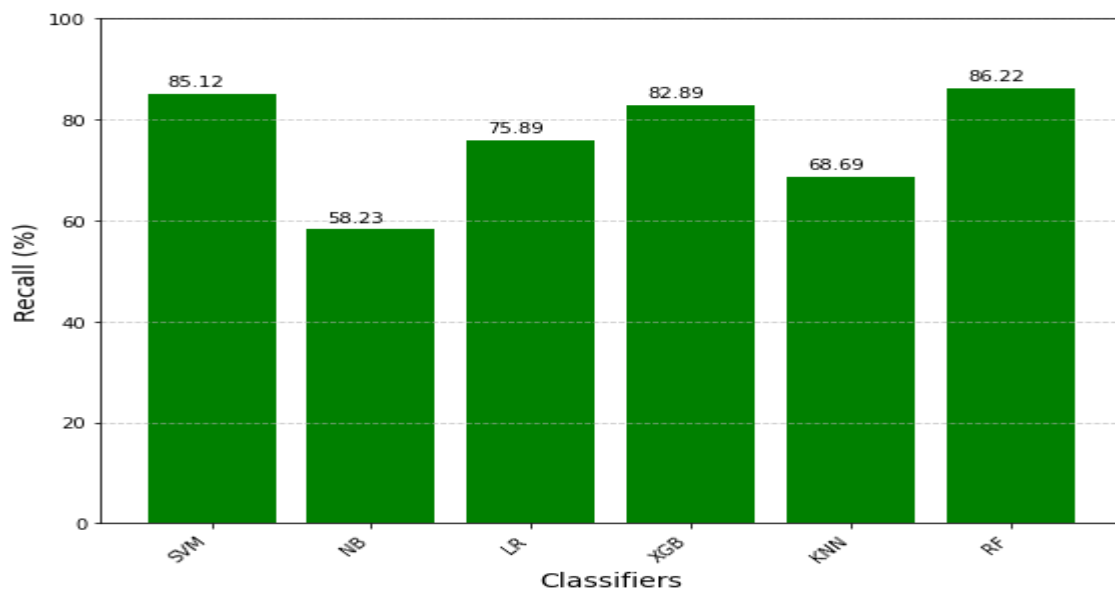


Fig. 8: Recall of machine learning classifiers.

The bar graph visualizes the accuracy percentages of F1-score of machine learning classifiers. The highest accuracy of F1-score is generated by the Random forest(RF) classifier (85.05%), followed closely by SVM (83.26%), Naive Bayes(NB) shows the lowest

F1-score (54.18%) while other classifiers like Logistic Regression(LR) (74.62%) and KNN (66.39%) show moderate performance. Each bar indicates a classifier's F1-score accuracy, with values displayed on top for clarity.

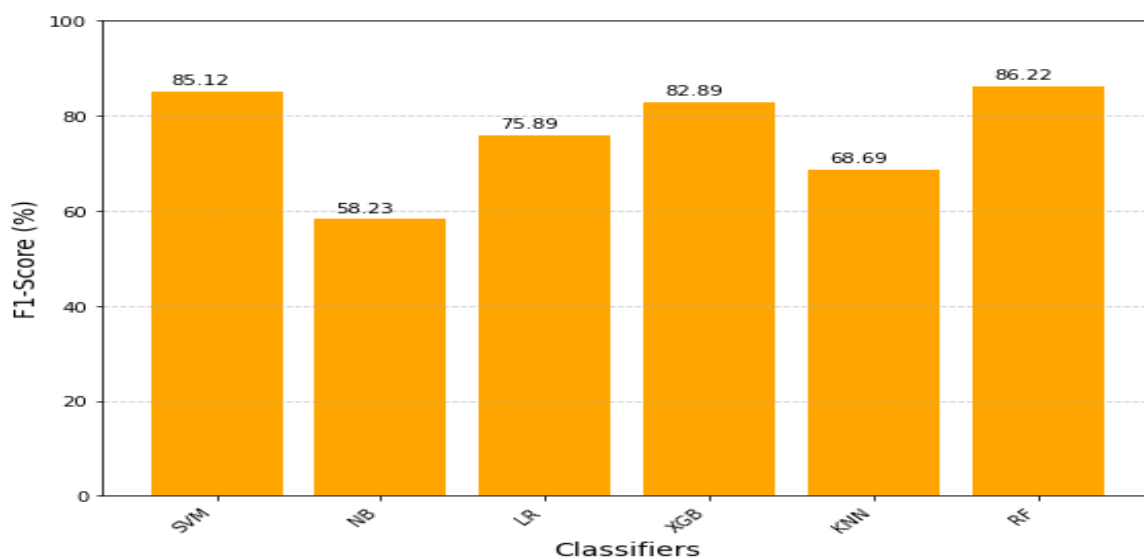


Fig. 8: F1-Score of machine learning classifiers.

Table 1. - Predicted Output by classifiers for unkonwen plant leaf image sample

Classifiers	Accuracy(%)	Precision(%)	Recall (%)	F1-Score (%)	Plant Type	Leaf	Leaf-HealthLabel
SVM	85.00	85.12	85.12	84.12	Arjun		Healthy
Naive Bayes	58.23	62.26	58.23	56.23	Bael		Unhealthy
Logistic Regression	75.98	75.89	75.89	75.45	Arjun		Healthy
KNN	68.79	69.26	68.79	66.00	Arjun		Healthy
Random Forest	86.32	86.22	86.22	86.22	Arjun		Healthy
XGBOOST	82.46	82.98	82.89	81.12	Arjun		Healthy

Table:1 shows the predicted plant leaf type and leaf health label for the pre-processed leaf image shown in Figure 5 . Leaf is correctly identified by the SVM, LR, KNN, RF and XGBoost classifiers respectively while

NB fails to Identify. The Random Forest technique has shown the highest accuracy compared to other classifiers.

Random Forest is ideal for work like disease identification in leaf datasets where variables like colour, texture, and shape are crucial since it uses decision trees to handle non-linear data successfully. By training each tree on random segments of the data, it avoids over-fitting and improves the model's performance on new, unpublished data. Random Forest is computationally efficient due to its capacity to handle several trees simultaneously, particularly for huge datasets. Random Forest is appropriate for unbalanced datasets since it manages class imbalance better than more straightforward models like Naive Bayes. Random Forest creates a strong ensemble model by integrating predictions from several decision trees, which can recognise a variety of patterns for precise disease detection.

CONCLUSION

Machine learning based applications are widely applied in different domains. Large dataset provides an opportunity to machine learning classifiers to become more accurate. A Random Forest classifier-based model with high accuracy (86.32%) and well accepted F1-score (86.22%) is presented. This model predicts leaf type and leaf health types simultaneously for a given unknown leaf image. It is beneficial on unknown leaf. More leaf species dataset would make model more robust.

Used well accepted SMVU dataset have 11 different types of leaf images, a larger number of images would also be more beneficial. Framework involves the application of deep learning models and use of transfer learning.

Extended form of conclusion

Machine learning-based applications are increasingly prevalent across diverse domains, offering innovative solutions to complex problems. This research demonstrates the potential of machine learning for medicinal plant classification and health prediction. By leveraging a Random Forest classifier, the proposed model achieves a commendable accuracy of 86.32% and an F1-score of 86.22%, underscoring its effectiveness and reliability. The model's ability to simultaneously predict both leaf type and health status for an unknown leaf image highlights its practical utility, particularly in scenarios requiring real-time or automated analysis.

The use of the SMVU dataset, comprising 11 distinct types of leaf images, has provided a solid foundation for training and evaluating the model. However, incorporating a more extensive and diverse dataset in future work could significantly enhance the model's robustness and generalizability. Furthermore, the integration of deep learning architectures and transfer learning methodologies has proven instrumental in improving feature extraction and classification accuracy. These techniques not only optimize the learning process but also reduce the dependency on extensive domain-specific expertise.

Future extensions of this research could explore the inclusion of additional medicinal plant species and health conditions, paving the way for broader

applicability in agriculture, medicine, and environmental monitoring. The framework demonstrates a scalable and adaptable approach, offering a promising avenue for further advancements in AI-driven plant analysis. Overall, this study lays the groundwork for developing intelligent systems that can revolutionize plant-based research and contribute meaningfully to health and agricultural sciences.

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